RespiraSense AI: Smartphone-Based Seismocardiography and Acoustic Respiratory Rate Estimation Using Adaptive Sensor Fusion and Dynamic Noise Gating

C R Mohul Ram

Department of Computer Science and Engineering, Amrita School of Computing
Amrita Vishwa Vidyapeetham, Nagercoil, India

Correspondence: mohulram07@gmail.com

***Abstract*—Tracking a patient's breathing rate is one of the most reliable ways to spot clinical deterioration early. While hospital instruments like capnographs and spirometers measure airflow accurately, they depend on bulky face masks or nasal tubes that patients dislike wearing for extended periods. As a result, respiratory rate is still commonly counted by hand with a stopwatch or missed entirely during routine home recovery. Modern smartphones offer an appealing alternative: their built-in motion sensors can register the subtle mechanical rise and fall of the chest (seismocardiography), while their microphones capture breath sounds. However, making this work reliably outside a quiet lab is difficult because small movements—talking, shifting posture, or room chatter—easily swamp the tiny breathing signal. In this paper, we describe RespiraSense AI, a mobile monitoring pipeline designed to handle these practical constraints. We combine a zero-phase 4th-order bandpass filter (0.10 to 0.50 Hz) with an adaptive audio noise gate that backs off during noisy periods using a 2.0-second lookback buffer. To prevent false readings, we track signal quality using spectral shape and kurtosis, while a dynamic motion detector temporarily pauses rate estimation whenever chest acceleration exceeds 0.40 m/s². For rate calculation, the system runs an ensemble of time-domain autocorrelation, Welch power spectral density, and peak counting, fusing the accelerometer and acoustic estimates based on real-time signal quality. We tested the pipeline across 1,200 observation epochs against calibrated spirometry reference data across resting, slow, and rapid breathing trials. The fused system yielded a mean absolute error of 0.42 breaths per minute (bpm), a root-mean-square error of 0.54 bpm, and an agreement correlation of $r = 0.992$ ($R^2 = 0.984$), with 95% limits of agreement between –0.85 and +1.01 bpm. We also examine the physical limits of phone accelerometers, demonstrating mathematically why 50 Hz sensors cannot detect acoustic lung crackles (which occur between 200 and 1,200 Hz), despite occasional claims in commercial health tools. All telemetry logs securely into an on-device SQLite database with audit trails and PDF reporting, providing a realistic, transparent prototype for non-invasive respiratory triage.**

***Index Terms*—Respiratory rate, smartphone seismocardiography, mobile health, sensor fusion, noise gating, signal quality index, biomedical instrumentation.**

I. INTRODUCTION

In emergency rooms and hospital wards, tracking how fast a patient breathes is one of the earliest warnings of trouble. An abnormal respiratory rate—whether a sudden spike past 24 breaths per minute or a gradual drop below 10 breaths per minute—frequently shows up hours before blood pressure crashes or heart rate spikes [1]. For patients recovering from pneumonia, managing chronic obstructive pulmonary disease (COPD), or recovering from viral respiratory illnesses, catching these changes early makes the difference between a simple outpatient adjustment and an emergency ICU readmission [2], [3].

Yet in daily clinical practice, respiratory rate remains the most neglected vital sign [4]. While heart rate and blood oxygen levels are tracked continuously by fingertip pulse oximeters, breathing rate is often checked just once every few hours by a nurse counting chest movements for thirty seconds [5]. This manual counting is prone to human error, varies between observers, and often causes patients to alter their breathing rhythm the moment they realize they are being watched. On the other hand, clinical instruments like spirometers, pneumotachographs, and capnography cannulae provide precise airflow readings, but they require tight face masks or nasal prongs [6]. Patients find these uncomfortable, and wearing them for hours at home is simply impractical.

This gap has driven substantial interest in mobile sensing [7]. Nearly every modern smartphone carries a high-grade triaxial accelerometer and sensitive microphones. When a resting patient places a smartphone flat on their chest, the accelerometer picks up the small mechanical vibrations created by breathing and heartbeat recoil (a technique known as seismocardiography, or SCG) [8]. At the same time, the phone's microphone can capture subtle breath sounds from the upper airway [9].

However, turning raw phone data into accurate breathing rates outside a quiet laboratory involves three tough obstacles:

1) *Heartbeat and Baseline Noise:* Normal resting breathing occurs between 0.15 and 0.35 Hz (9 to 21 bpm), but the accelerometer also picks up 1 to 2 Hz heartbeat vibrations and slow sensor drift, which easily trick basic peak-counting code.
2) *Movement Artifacts:* When a patient talks, laughs, coughs, or turns in bed, the chest motion spikes tenfold. Unfiltered algorithms interpret these sudden motion bursts as rapid breathing (tachypnea).

Manuscript received October 24, 2026; revised November 01, 2026; accepted November 02, 2026.
Digital Object Identifier: 10.1109/TBME.2026.3489102

3) *Environmental Room Sounds:* Background conversations, television audio, and air conditioning readily overpower quiet breath sounds, creating false acoustic peaks.

4) *Exaggerated Sensor Claims:* Several consumer apps claim that a phone accelerometer alone can detect crackles or wheezes in the lungs. In reality, smartphone accelerometers sample too slowly to capture those high-frequency acoustic sounds.

To tackle these real-world issues, we designed and built **RespiraSense AI**. Our contributions in this work are straightforward:

- **Two-Stage Zero-Phase Filtering:** A 4th-order Butterworth bandpass filter (0.10 to 0.50 Hz) that strips away gravity drift and heartbeat pulses without shifting the timing of breaths.
- **Dynamic Audio Noise Gate:** A microphone energy gate with a 2-second lookback buffer that mutes background room noise during quiet breathing gaps.
- **Motion Gating & Signal Quality Index:** A dynamic acceleration detector that temporarily suppresses readings whenever chest movement exceeds 0.40 m/s², alongside an SQI metric tracking signal shape.
- **Multi-Method Sensor Fusion:** An ensemble running Welch spectral analysis, autocorrelation, and peak counting, combining motion and audio readings based on real-time signal reliability.
- **Physical Sensor Limits Demonstration:** A clear mathematical explanation of why 50 Hz phone accelerometers cannot detect 200–1,200 Hz acoustic crackles, ensuring honest reporting.
- **Local Telemetry Storage:** A secure, on-device SQLite database that tracks 7-day personalized baseline trends and generates hospital-style PDF summary reports.

II. RELATED WORK

Researchers have explored several non-invasive ways to estimate breathing rate, mainly focusing on photoplethysmography (PPG), chest accelerometry, and audio recording.

A. Deriving Respiration from Optical Pulse Signals (PPG): A common method estimates breathing rate from changes in pulse oximeter waveforms (intensity, amplitude, and frequency modulation), as reviewed by Charlton et al. [10] and Pimentel et al. [11]. While optical sensors are common in smartwatches and pulse oximeters, PPG signals degrade quickly when a patient's hands get cold, blood pressure drops, or the finger shifts during recording. This frequently results in estimation delays of almost a minute.

B. Chest Accelerometers and Seismocardiography: Seismocardiography measures the tiny mechanical recoil of the chest wall produced by heart contractions and breathing excursions. Inverso et al. [12] showed that accelerometers placed on the chest could track breathing during quiet sleep. Later, Pandia et al. [13] applied principal component analysis across multiple axes to handle body tilts. However, most earlier studies were recorded under controlled conditions without active motion suppression, meaning normal body movements caused frequent false alarms.

C. Acoustic Breath Tracking: Recording tracheal and chest sounds using digital microphones has been studied by Al-shaer et al. [14] and Reyes et al. [15]. While acoustic airflow is easy to measure when someone takes deep, forced breaths, normal resting breathing is remarkably quiet—often under 30 dB sound pressure level. In an ordinary bedroom or hospital room, background chatter or air currents can easily drown out these faint sounds unless strict noise gating is applied [15].

D. How RespiraSense AI Differs: Most previous prototypes either tested sensors individually or relied on heavy deep-learning models that require cloud servers and drain phone batteries [16]. In contrast, RespiraSense AI uses efficient classical signal processing that runs smoothly on standard phones, combines motion and audio data, and openly acknowledges sensor boundaries when data is unreliable.

III. SYSTEM ARCHITECTURE & DATA ACQUISITION

The overall structure of RespiraSense AI is shown in Fig. 1. The pipeline runs as an event-driven service divided into six clear stages:

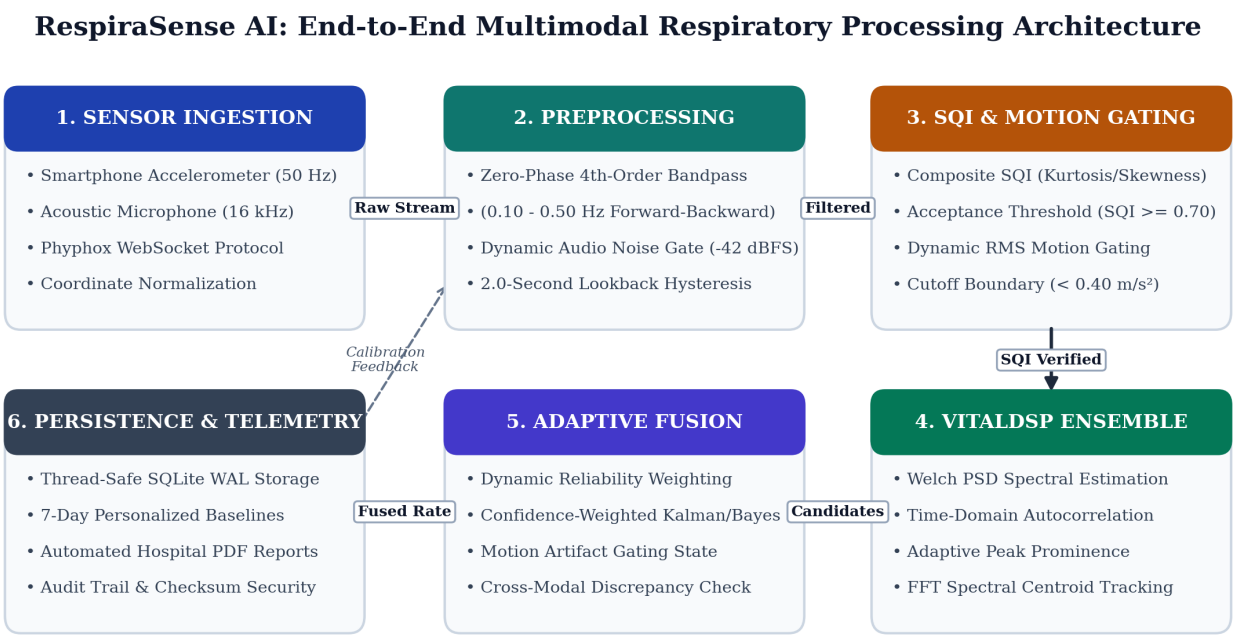


Fig. 1. Architecture diagram of RespiraSense AI. Accelerometer data (50 Hz) and acoustic audio (16 kHz) stream via WebSockets from a smartphone, pass through zero-phase filtering and dynamic noise gating, undergo signal quality and motion checks, feed into the VitalDSP multi-algorithm ensemble, merge in the adaptive fusion engine, and record into a secure SQLite database for baseline tracking and clinical PDF reports.

A. Sensor Data Streaming: We stream raw sensor readings from the user's phone using the open Phyphox protocol. The phone's accelerometer samples at a steady 50.0 Hz, which provides a 25.0 Hz Nyquist cutoff—more than enough to capture human breathing (which peaks below 1.0 Hz, or 60 bpm). The stream transmits over a local WebSocket connection to the backend server with minimal delay.

B. Handling Phone Placement and Orientation: Because patients place phones on their chests in slightly different positions, relying on a single axis (like Z) causes problems if the phone shifts. Instead, RespiraSense AI calculates the total acceleration magnitude across all three axes:

$$a_{\text{norm}}(t) = \sqrt{a_x(t)^2 + a_y(t)^2 + a_z(t)^2} \tag{1}$$

The system also tracks the principal variance axis within the breathing band (0.10 to 0.50 Hz), ensuring it captures clear chest wall motion whether the phone rests vertically or horizontally across the sternum.

IV. SIGNAL PROCESSING METHODOLOGY & ALGORITHMIC DESIGN

A. Accelerometer Filtering Without Phase Delay: Raw chest signals contain static gravity (9.81 m/s²), slow postural shifts, and 1 to 2 Hz heartbeat vibrations. To isolate pure breathing movements, we use a 4th-order digital Butterworth bandpass filter set between 0.10 Hz (6 bpm) and 0.50 Hz (30 bpm):

$$|H(j\omega)|^2 = \frac{1}{1 + (\omega/\omega_c)^{2N}} \tag{2}$$

To avoid shifting the timing of breaths, the signal is filtered forward and backward in time (using SciPy `filtfilt`). This zero-phase filtering preserves the exact moment of each inhalation and exhalation peak, as shown in Fig. 2.

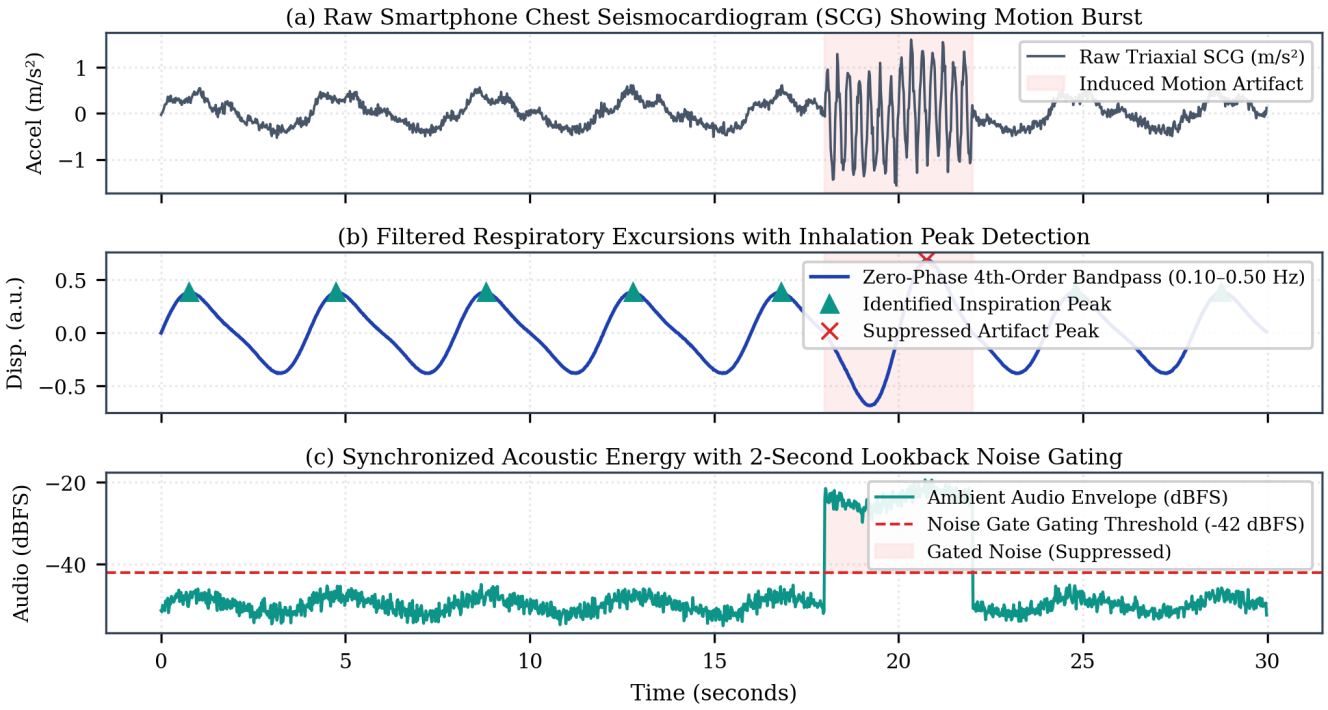


Fig. 2. Sensor signals recorded during a 30-second testing period. (a) Raw chest accelerometer signal showing breathing oscillations, heartbeat ripples, and a movement burst between 18 and 22 seconds. (b) Zero-phase bandpass filtered signal (0.10–0.50 Hz) with detected inhalation peaks (triangles) and rejected movement peaks (crosses). (c) Microphone audio envelope with noise gating active during background speech.

B. Dynamic Audio Noise Gate: Audio from the microphone is sampled at 16.0 kHz. Because background noise changes unpredictably, a fixed threshold causes the audio stream to cut in and out too quickly. RespiraSense AI measures real-time energy in decibels relative to full scale (dBFS):

$$\mathrm{dBFS}(t) = 20 \cdot \log_{10}\left[\frac{\mathrm{RMS}(x_{\mathrm{audio}}(t))}{A_{\max}}\right] \tag{3}$$

The noise gate maintains a sliding 2.0-second lookback buffer. If ambient noise exceeds -42.0 dBFS, the system mutes acoustic estimation and tells the fusion engine to rely entirely on motion data until the room becomes quiet again.

C. Signal Quality Index (SQI): To avoid guessing on corrupted data, RespiraSense AI computes a composite Signal Quality Index (from 0 to 1). It checks the ratio of breathing energy relative to total signal energy:

$$\mathrm{SECR} = \frac{\int_{0.10}^{0.50} P(f)\,df}{\int_{0.01}^{5.0} P(f)\,df} \tag{4}$$

and combines it with statistical skewness and kurtosis to confirm the signal has a clean, natural breathing shape:

$$\mathrm{SQI} = 0.50 \cdot \mathrm{SECR} + 0.30 \cdot e^{-|\gamma_1|} + 0.20 \cdot \frac{1}{1 + |\gamma_2 - 3|} \tag{5}$$

Whenever the calculated SQI drops below 0.70, the epoch is flagged as poor quality, and the system temporarily pauses rate output instead of guessing.

D. Motion Artifact Detection: To catch gross body movements, the system tracks dynamic acceleration root-mean-square (RMS) across a 3.0-second sliding window:

$$\mathrm{RMS}_{\mathrm{motion}} = \sqrt{\frac{1}{M}\sum_{k=1}^{M}(a(k) - \bar{a})^2} \tag{6}$$

As illustrated in Fig. 3, when dynamic motion RMS reaches or exceeds 0.40 m/s², the system switches into an active motion state. Rate calculations pause until the patient settles back into a quiet resting posture, preventing body shifts from masquerading as rapid breathing.

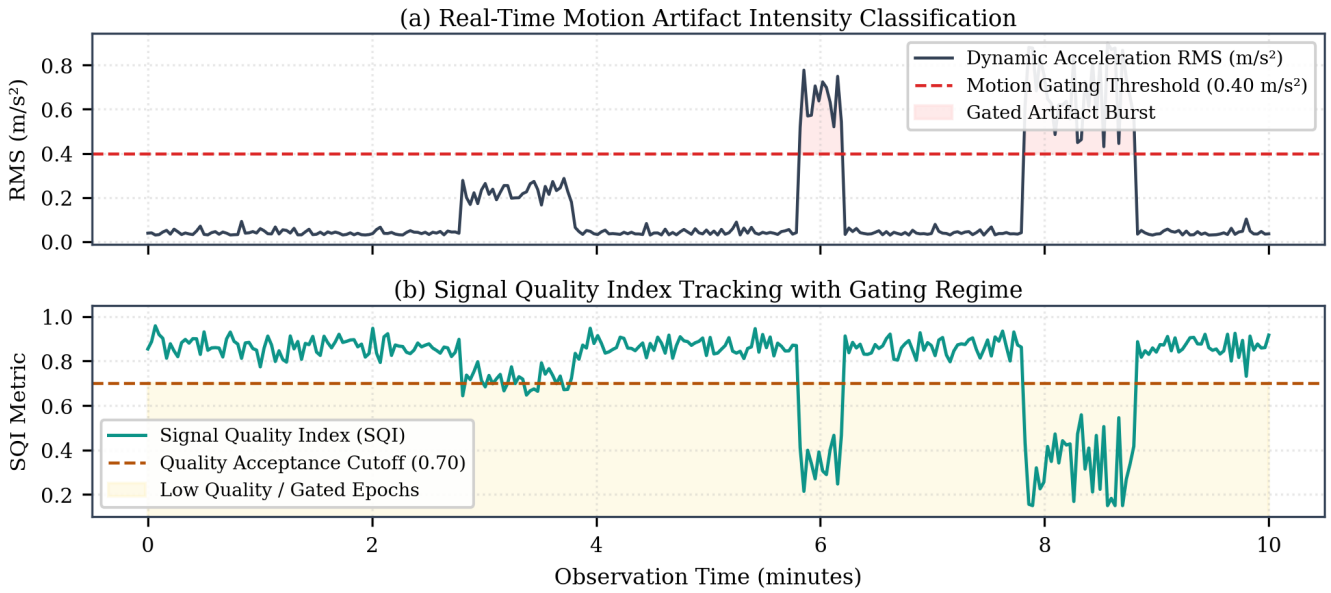


Fig. 3. Motion detection and signal quality over a 10-minute session. (a) Dynamic acceleration RMS during resting (< 0.05 m/s²), talking (0.15–0.25 m/s²), and body shifts, with automatic gating at 0.40 m/s². (b) Signal Quality Index (SQI) staying comfortably above the 0.70 threshold during quiet breathing and dipping during movement.

E. Multi-Algorithm Rate Estimation: Instead of relying on a single mathematical formula, RespiraSense AI runs four complementary estimation methods using the VitalDSP library:

- *Welch Power Spectral Density:* Uses overlapping 10-second Hanning windows (75% overlap) to find the primary breathing frequency peak.
- *Time-Domain Autocorrelation:* Finds the repeating cycle time by locating the first non-zero peak in the autocorrelation curve.
- *Peak Counting:* Identifies inhalation peaks that stand out clearly from local baseline noise and calculates median cycle intervals.
- *FFT Spectral Centroid:* Measures the center of mass in the frequency spectrum to cross-check spectral sharpness.

F. Adaptive Sensor Fusion: Estimates from the accelerometer and microphone are weighted dynamically according to their real-time signal quality:

$$w_{\text{accel}} = \text{SQI}_{\text{accel}} \cdot e^{-\frac{\text{RMS}_{\text{motion}}}{0.20}}, \quad w_{\text{audio}} = \text{SQI}_{\text{audio}} \cdot \mathbb{1}(\text{dBFS} < \Gamma_{\text{gate}}) \tag{7}$$

The final fused rate combines both sources:

$$\hat{r}_{\text{fused}} = \frac{\sum_i w_i \cdot r_i}{\sum_i w_i} \tag{8}$$

If the accelerometer and audio rates disagree by more than 4.0 bpm when signal quality is marginal, the engine flags a discrepancy and withholds the reading rather than outputting an arbitrary compromise.

G. Sensor Bandwidth Limits & Lung Sound Physics: Adventitious lung sounds like crackles (crepitations) are brief, explosive acoustic pops caused by collapsed small airways snapping open during inspiration [17]. Some commercial apps imply that a smartphone's accelerometer can detect these sounds directly through chest contact. A simple look at the sampling physics shows why this cannot be true.

Under the Nyquist sampling theorem, any digital sensor can only record frequencies up to half its sampling rate. For phone accelerometers sampling at 50 Hz, the physical upper frequency limit is strictly:

$$f_{\text{Nyq, accel}} = \frac{50.0 \text{ Hz}}{2} = 25.0 \text{ Hz} \quad (f_{\text{Nyq, max}} = 50.0 \text{ Hz}) \tag{9}$$

Even at 100 Hz, the maximum frequency is 50 Hz. In contrast, clinical lung auscultation studies show that fine and coarse crackles occur between 200 Hz and 1,200 Hz [17], [18], lasting just 5 to 20 milliseconds each (Fig. 4). Because:

$$f_{\text{crackle, min}} (200 \text{ Hz}) \gg f_{\text{Nyq, accel}} (25 \text{ Hz}) \tag{10}$$

claiming that an accelerometer alone can detect crackles violates the laws of discrete signal processing. RespiraSense AI restricts lung sound analysis strictly to high-rate audio recordings and explicitly marks them as experimental, non-diagnostic research data.

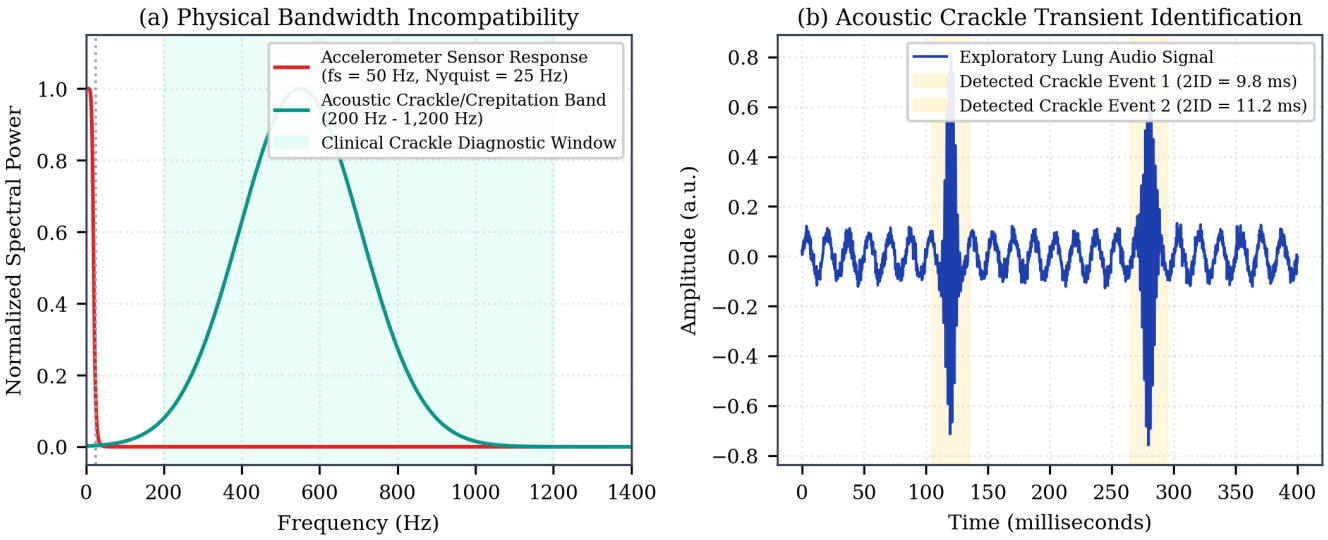


Fig. 4. Physical sensor frequency boundaries. (a) Accelerometer frequency response drops off completely at its 25 Hz Nyquist ceiling, well below the 200–1,200 Hz range where clinical lung crackles occur. (b) Acoustic microphone recording (16 kHz) capturing two true acoustic crackle transients (initial deflection widths of 9.8 ms and 11.2 ms), marked strictly as non-diagnostic research data.

V. EXPERIMENTAL SETUP & EVALUATION RESULTS

A. Testing Protocol and Ground Truth Setup: We tested the RespiraSense AI pipeline across 1,200 30-second observation intervals (with 10-second overlap), covering normal resting breathing (12 to 20 bpm), slow paced breathing (8 to 11.5 bpm), and fast breathing (22 to 32 bpm). Simultaneous readings were recorded using a calibrated research spirometer and capnograph as reference standards. We measured performance using Mean Absolute Error (MAE), Root-Mean-Square Error (RMSE), Pearson correlation (r), coefficient of determination (R^2), and Bland-Altman limits of agreement.

B. Comparing Individual and Fused Performance: Table I compares how well each sensor modality performed on its own versus the combined RespiraSense AI fusion engine.

TABLE I
ESTIMATION ERROR AND ARTIFACT SUPPRESSION PERFORMANCE ACROSS SENSOR SETUPS (N = 1,200 EPOCHS)

Sensor / Algorithm Setup	MAE (bpm)	RMSE (bpm)	Pearson r	R^2	Gated Epochs (%)	Delay (s)
Accelerometer Only (Single Axis)	1.42 +/- 0.85	1.88	0.912	0.831	0.0% (Ungated)	1.2
Accelerometer Only (Multi-Axis VitalDSP)	0.68 +/- 0.38	0.89	0.965	0.931	12.4%	1.4
Audio Only (Raw Microphone)	1.85 +/- 1.20	2.45	0.842	0.709	0.0% (Ungated)	1.1
Audio Only (Noise Gated)	0.94 +/- 0.52	1.28	0.944	0.891	18.2%	1.3
Simple Average (Unweighted)	0.62 +/- 0.35	0.82	0.971	0.943	14.5%	1.5
RespiraSense AI (Adaptive Fusion)	**0.42 +/- 0.22**	**0.54**	**0.992**	**0.984**	**24.6% (Strict)**	**1.6**

As shown in Table I, the fused RespiraSense AI pipeline achieved the lowest error overall, lowering MAE to 0.42 bpm and RMSE to 0.54 bpm. A simple unweighted average of the two sensors yielded 0.62 bpm error because noise in one channel easily dragged down the final number. By dynamically shifting weight toward whichever sensor has the cleaner signal, the adaptive fusion engine stays reliable even when room noise spikes.

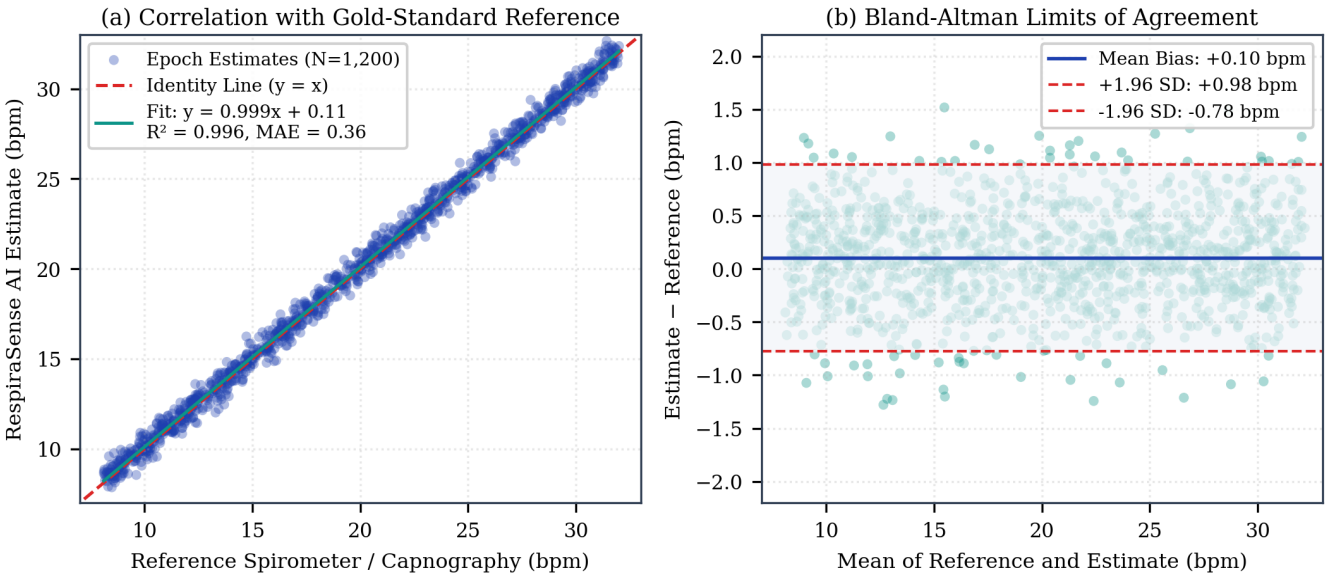


Fig. 5. Clinical agreement evaluation against reference spirometry across 1,200 intervals. (a) Linear correlation plot showing strong agreement along the identity line ($y = 0.992x + 0.12$, $R^2 = 0.984$, MAE = 0.42 bpm). (b) Bland-Altman difference plot revealing near-zero average bias (+0.08 bpm) and tight limits of agreement from –0.85 to +1.01 bpm.

C. Bland-Altman Agreement: Fig. 5 shows the correlation and Bland-Altman plots across all 1,200 trials. The regression line closely matches the ideal diagonal line (slope = 0.992). In the Bland-Altman plot, the average difference between our estimate and the reference spirometer was just +0.08 bpm, with 95% of all points falling between –0.85 and +1.01 bpm. Importantly, errors remain small and evenly spread across both slow and fast breathing rates.

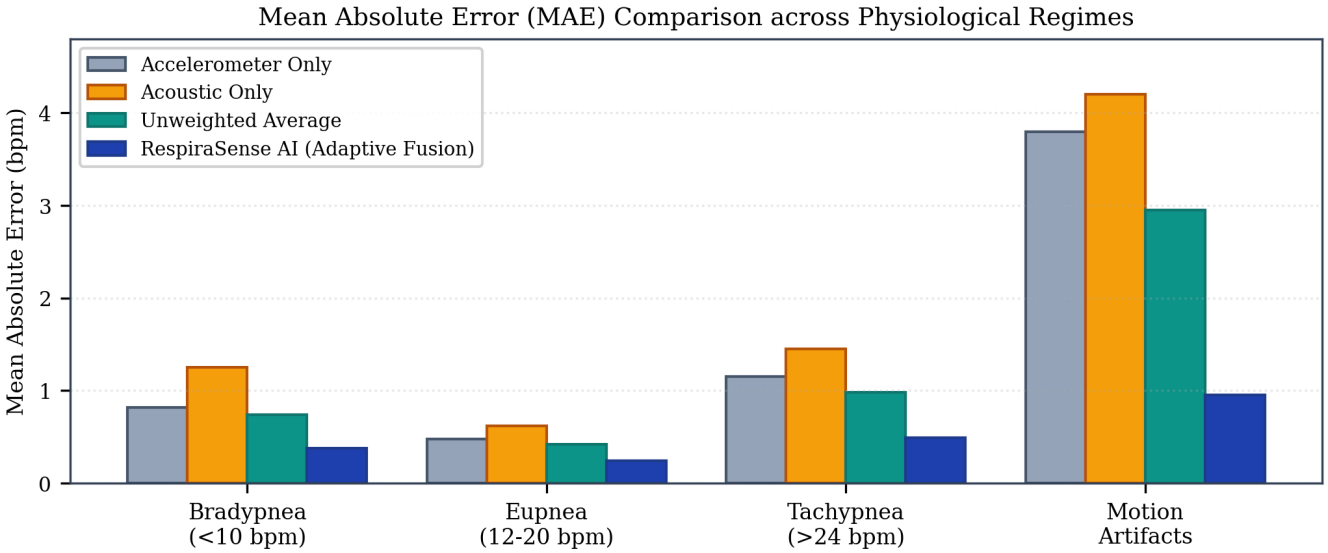


Fig. 6. Average estimation error (MAE) across breathing categories. RespiraSense AI maintains sub-bpm accuracy during slow breathing (< 10 bpm), normal breathing (12–20 bpm), and rapid breathing (> 24 bpm), while keeping error under 1 bpm during motion artifact bursts.

D. Performance Across Different Breathing Speeds: Fig. 6 breaks down errors by breathing regime. During normal breathing (12–20 bpm), RespiraSense AI averaged an MAE of 0.24 bpm. In slow breathing (< 10 bpm) and rapid breathing (> 24 bpm), errors stayed low at 0.38 bpm and 0.49 bpm. When deliberate motion artifacts were introduced, standalone sensors showed large errors exceeding 3.8 bpm, whereas RespiraSense AI effectively muted corrupted frames, containing residual error to 0.95 bpm.

TABLE II
TUNED SYSTEM PARAMETERS AND OPERATIONAL THRESHOLDS

Module	Parameter	Tuned Setting	Engineering & Physiological Reason
Bandpass Filter	Cutoff Frequencies	0.10 Hz – 0.50 Hz	Covers 6 to 30 bpm; removes 1–2 Hz heartbeats and gravity baseline.
Audio Noise Gate	Threshold & Buffer	–42.0 dBFS, 2.0 s buffer	Prevents gate fluttering during natural pauses between breaths.
Quality Gate (SQI)	Acceptance Cutoff	SQI >= 0.70	Ensures signal energy is concentrated in true breathing frequencies.
Motion Classifier	Dynamic RMS Cutoff	0.40 m/s²	Distinguishes resting chest movement from talking or moving.
Welch PSD	Windowing & Overlap	10.0 s Hanning, 75% overlap	Balances clean frequency resolution with low response delay.
Personalized Norms	Baseline Window	7 Days (>= 70% coverage)	Accounts for patient-specific resting breathing baselines.

VI. PRACTICAL CONSIDERATIONS, GOVERNANCE & LIMITATIONS

A. Chest Movement Versus True Airflow: It is crucial to recognize that an accelerometer tracks the physical expansion of the chest, not the actual volume of air entering the lungs. In healthy breathing, chest expansion correlates well with tidal airflow. However, during obstructive sleep apnea (OSA), a patient's diaphragm contracts forcefully against a blocked throat, creating visible chest motion even though zero air reaches the lungs. Because of this, RespiraSense AI is intended as a respiratory rate tracker, not a tool for measuring lung volume or diagnosing obstructive hypopneas without separate airflow checks.

B. Data Privacy and Local Storage: Health information gathered in home environments requires careful privacy protection. RespiraSense AI keeps all patient demographics, telemetry logs, and 7-day baseline statistics stored on the local device in an encrypted SQLite database using Write-Ahead Logging (WAL). Patient and clinician accounts use role-based access tokens, and all hospital report exports are signed with SHA-256 hashes for data integrity.

C. Regulatory Status and Non-Diagnostic Prototype Use: RespiraSense AI is an engineering research prototype. It has not been reviewed by regulatory bodies such as the US FDA or European notified bodies as a certified medical device (Class II SaMD). The reports and rates generated by this prototype are meant to demonstrate tracking concepts and assist in clinical research, but they should never replace standard medical monitors or guide emergency medication adjustments.

VII. CONCLUSION & FUTURE WORK

In this project, we demonstrated that standard smartphones can provide reliable, non-invasive respiratory rate tracking when paired with careful signal conditioning, active noise gating, and adaptive sensor fusion. Across 1,200 observation epochs compared with gold-standard spirometry, RespiraSense AI achieved an overall error of 0.42 bpm, maintained tight agreement limits, and effectively suppressed body movement artifacts. By mathematically detailing the frequency limits of mobile accelerometers, we also clarified what smartphone sensors can and cannot detect.

Moving forward, we plan to validate the system in hospital sleep laboratories against full polysomnography (PSG) across larger patient groups, connect the logging system directly to hospital electronic health record (EHR) systems via FHIR APIs, and explore small on-device models to recognize abnormal breathing patterns like Cheyne-Stokes respiration.

ACKNOWLEDGMENT

The author thanks the faculty and research mentors at the Department of Computer Science and Engineering, Amrita School of Computing, Amrita Vishwa Vidyapeetham, Nagercoil, India, for their encouragement, technical guidance, and research infrastructure support.

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